“see” the ball, the paddle, the blocks, and to recognize the direction of the ball. That’s pretty amazing given that it’s only being given raw pixel data. And even more interesting is that the learned high-level features may be transferable to other games or environments.

Deep learning is the secret sauce that makes all the recent successes in RL possible. No other class of algorithms has demonstrated the representational power, efficiency, and flexibility of deep neural networks. Moreover, neural networks are actually fairly simple!

1.7. Our didactic tool: String diagrams

The fundamental concepts of RL have been well-established for decades, but the field is moving very quickly, so any particular new result could soon be out of date. That’s why this book focuses on teaching skills, not details with short half-lives. We do cover some recent advances in the field that will surely be supplanted in the not too distant future, but we do so only to build new skills, not because the particular topic we’re covering is necessarily a time-tested technique. We’re confident that even if some of our examples become dated, the skills you learn will not, and you’ll be prepared to tackle RL problems for a long time to come.

Moreover, RL is a huge field with a lot to learn. We can’t possibly hope to cover all of it in this book. Rather than be an exhaustive RL reference or comprehensive course, our goal is to teach you the foundations of RL and to sample a few of the most exciting recent developments in the field. We expect that you will be able to take what you’ve learned here and easily get up to speed in the many other areas of RL. Plus, we have a section in chapter 11 that gives you a roadmap of areas you might want to check out after finishing this book.

This book is focused on teaching well, but also rigorously. Reinforcement learning and deep learning are both fundamentally mathematical. If you read any primary research articles in these fields, you will encounter potentially unfamiliar mathematical notations and equations. Mathematics allows us to make precise statements about what’s true and how things are